TABLE OF CONTENT

1. Introduction

This project implements a Public Health Clinic Records System aligned with SDG 3: Good Health and Well-being. The aim is to translate the Physical Data Model from Assignment 1 into a fully functional relational database using MySQL (XAMPP). The project demonstrates practical skills in database design, implementation, data manipulation, SQL queries, and user management.

2. Refined Physical Data Model

The refined model consists of five entities:

- Patients (patient_id, name, gender, dob, address, phone)

- Doctors (doctor_id, name, specialization, phone)

- Appointments (appointmentid, patientid, doctor_id, date, time, status)

- Treatments (treatmentid, appointmentid, description, cost)

- Payments (paymentid, treatmentid, amount, date)

ER DIAGRAM STRUCTURE

Patients

patient_id (PK)

name

gender

dob

address

phone

Doctors

doctor_id (PK)

names

pecialization

phone

Appointments

appointment_id (PK)

patient_id (FK)

doctor_id (FK)

date

time

status

Treatments

treatment_id (PK)

appointment_id (FK)

description

cost

Payments

payment_id (PK)

treatment_id (FK)

amount

date

Relationships:

- One patient → many appointments

- One doctor → many appointments

- One appointment → many treatments

- One treatment → one payment

3. Design Decisions

- Data types: VARCHAR for names/addresses, DATE for dob and appointment dates, DECIMAL for costs/payments.

- Constraints: Primary keys ensure uniqueness; foreign keys enforce relationships; NOT NULL ensures required fields; ENUM used for gender and appointment status.

- Relationships: Designed to reflect real- world clinic workflows (patients book appointments with doctors, appointments generate treatments, treatments require payments).

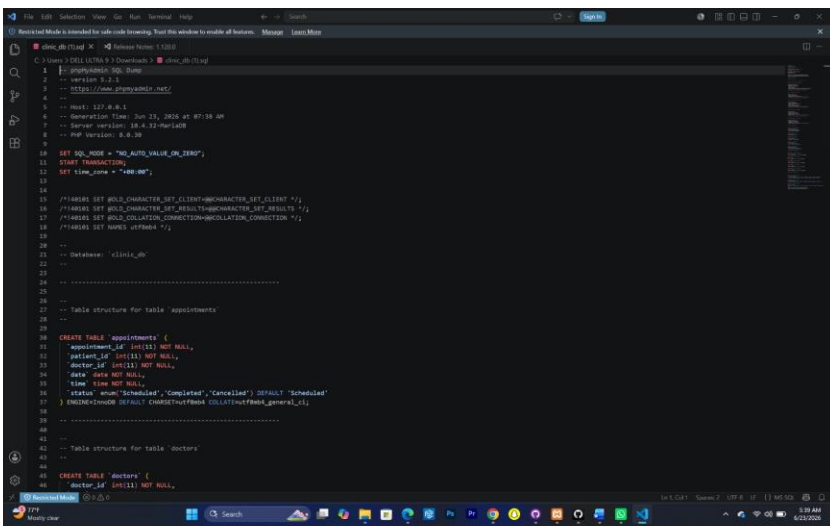
4. SQL Implementation

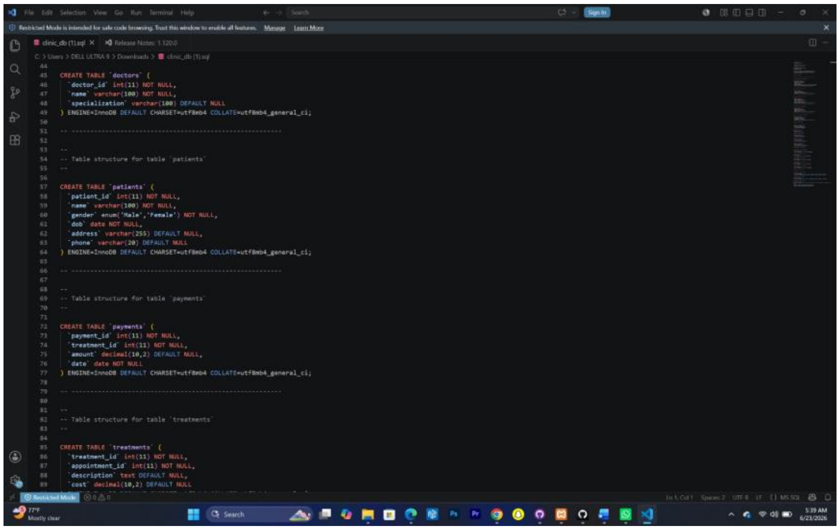
- Database Creation: CREATE DATABASE clinic_db;

- Tables: Patients, Doctors, Appointments, Treatments, Payments with PK/FK constraints.

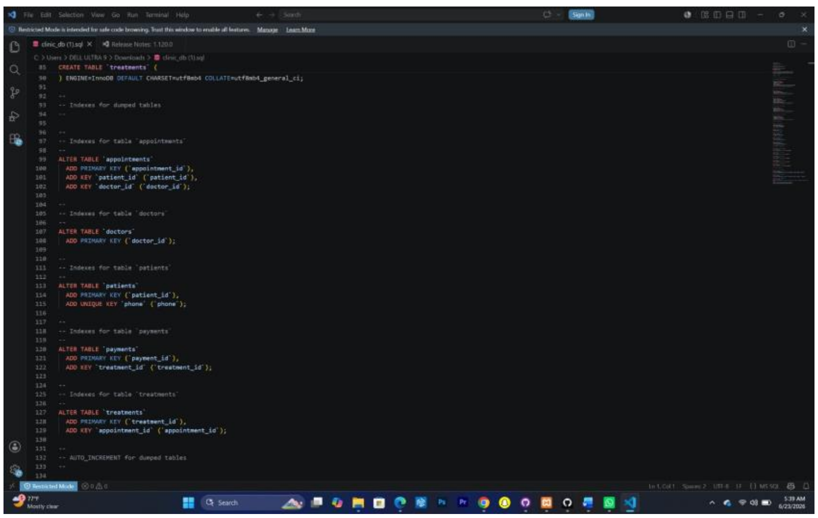
- Sample Data: Inserted meaningful records for patients, doctors, appointments, treatments, and payments.

- Update/Delete: Demonstrated modifying appointment status and deleting doctors after removing dependent appointments.

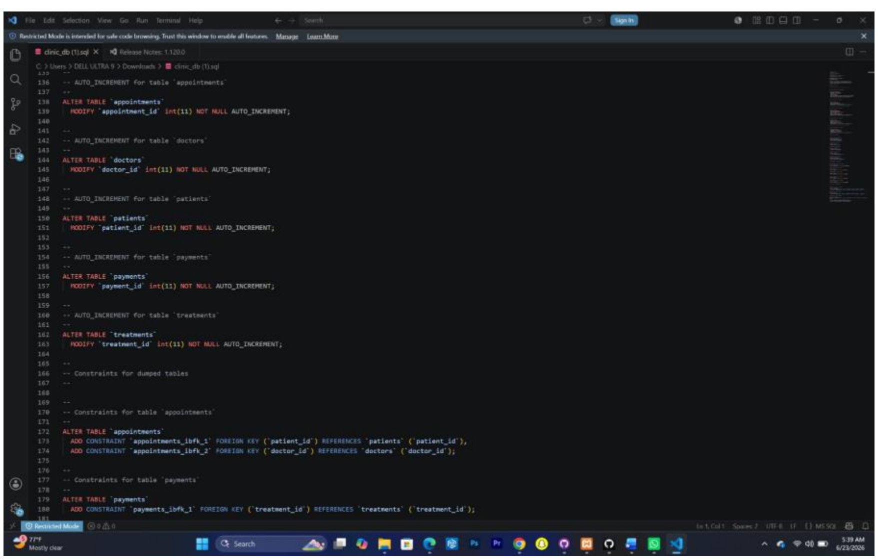




clinic_db (1).sql
CREATE TABLE `doctors` (
`doctor_id` int(11) NOT NULL,
`name` varchar(100) NOT NULL,
`specialization` varchar(100) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
-- --
--
-- Table structure for table `patients`
--
CREATE TABLE `patients` (
`patient_id` int(11) NOT NULL,
`name` varchar(100) NOT NULL,
`gender` enum('Male','Female') NOT NULL,
`dob` date NOT NULL,
`address` varchar(255) DEFAULT NULL,
`phone` varchar(20) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
-- --
--
-- Table structure for table `payments`
--
CREATE TABLE `payments` (
`payment_id` int(11) NOT NULL,
`treatment_id` int(11) NOT NULL,
`amount` decimal(10,2) DEFAULT NULL,
`date` date NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
-- --
--
-- Table structure for table `treatments`
--
CREATE TABLE `treatments` (
`treatment_id` int(11) NOT NULL,
`appointment_id` int(11) NOT NULL,
`description` text DEFAULT NULL,
`cost` decimal(10,2) DEFAULT NULL



clinic_db (1).sql
CREATE TABLE `treatments` (
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
--
-- Indexes for dumped tables
--
--
-- Indexes for table `appointments`
--
ALTER TABLE `appointments`
ADD PRIMARY KEY (`appointment_id`),
ADD KEY `patient_id` (`patient_id`),
ADD KEY `doctor_id` (`doctor_id`);
--
-- Indexes for table `doctors`
--
ALTER TABLE `doctors`
ADD PRIMARY KEY (`doctor_id`);
--
-- Indexes for table `patients`
--
ALTER TABLE `patients`
ADD PRIMARY KEY (`patient_id`),
ADD UNIQUE KEY `phone` (`phone`);
--
-- Indexes for table `payments`
--
ALTER TABLE `payments`
ADD PRIMARY KEY (`payment_id`),
ADD KEY `treatment_id` (`treatment_id`);
--
-- Indexes for table `treatments`
--
ALTER TABLE `treatments`
ADD PRIMARY KEY (`treatment_id`),
ADD KEY `appointment_id` (`appointment_id`);
--
-- AUTO_INCREMENT for dumped tables
--



clinic_db (1).sql
-- AUTO_INCREMENT for table `appointments`
--
ALTER TABLE `appointments`
MODIFY `appointment_id` int(11) NOT NULL AUTO_INCREMENT;
--
-- AUTO_INCREMENT for table `doctors`
--
ALTER TABLE `doctors`
MODIFY `doctor_id` int(11) NOT NULL AUTO_INCREMENT;
--
-- AUTO_INCREMENT for table `patients`
--
ALTER TABLE `patients`
MODIFY `patient_id` int(11) NOT NULL AUTO_INCREMENT;
--
-- AUTO_INCREMENT for table `payments`
--
ALTER TABLE `payments`
MODIFY `payment_id` int(11) NOT NULL AUTO_INCREMENT;
--
-- AUTO_INCREMENT for table `treatments`
--
ALTER TABLE `treatments`
MODIFY `treatment_id` int(11) NOT NULL AUTO_INCREMENT;
--
-- Constraints for dumped tables
--
--
-- Constraints for table `appointments`
--
ALTER TABLE `appointments`
ADD CONSTRAINT `appointments_ibfk_1` FOREIGN KEY (`patient_id`) REFERENCES `patients` (`patient_id`),
ADD CONSTRAINT `appointments_ibfk_2` FOREIGN KEY (`doctor_id`) REFERENCES `doctors` (`doctor_id`);
--
-- Constraints for table `payments`
--
ALTER TABLE `payments`
ADD CONSTRAINT `payments_ibfk_1` FOREIGN KEY (`treatment_id`) REFERENCES `treatments` (`treatment_id`);

5. SQL Queries

Examples include:

- Total treatments cost per patient

- Total payments made toward a treatment

- Patients with outstanding balances

- Queries using WHERE, ORDER BY, SUM, COUNT, AVG, and LIMIT.

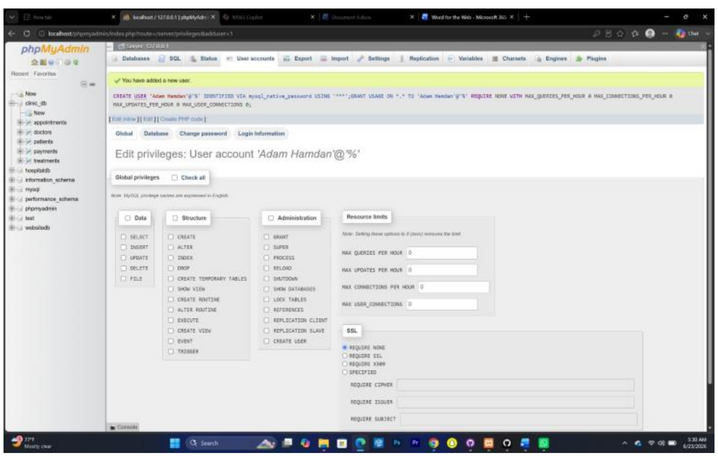
6. User Management

- Created accounts for group members.

- Granted privileges (SELECT, INSERT, UPDATE, DELETE).

- Changed passwords.

- Explained importance of database security (protecting patient data, preventing unauthorized access).



7. Conclusion

This project successfully demonstrates the design and implementation of a relational database system for a public health clinic. It covers all aspects of database management: modeling, implementation, data manipulation, queries, and user security. The project aligns with SDG 3 by supporting efficient healthcare record management in Sierra Leone.